HDOC Day-55

Question: Write a C program using pointers and nested loops to print the maximum element of each contiguous window of size k, separated by spaces. Print -1 for invalid k.

C Program:

```
#include <stdio.h>

void window_maximum(int *arr, int size, int k) {
    if (k <= 0 || k > size) {
        printf("-1\n");
        return;
    }

    for (int i = 0; i <= size - k; i++) {
        int max = arr[i];
        for (int j = i + 1; j < i + k; j++) {
            if (arr[j] > max)
                max = arr[j];
        }
        if (i > 0) printf(" ");
        printf("%d", max);
    }
    printf("\n");
}

int main(void) {
    int size, k;
    scanf("%d", &size);
    if (size <= 0) return 0;
    int arr[size];
    for (int i = 0; i < size; i++)
        scanf("%d", &arr[i]);
    scanf("%d", &k);
    window_maximum(arr, size, k);
    return 0;
}
```

Sample Input and Output:

```
Input:
8
1 3 -1 -3 5 3 6 7
3

Output:
3 3 5 5 6 7
```